Costs of Time

Tyler Jacobson

May 15, 2026

Agenda

- Stylized Model
- Entitlement Pipeline: Facts + Lessons
- Entitlements: Avoiding political uncertainty

Did see suggestive evidence in NYC that experience reduces permit withdrawal rates (6% for first project 2% for tenth project)

Entitlement Intuition

Risk of Failure

- Proposing a development project involves building a coalition of size Q: scales with size.
- Each member of the coalition succeeds at its task with probability:

$$\delta^T$$

- More time between when the coalition forms and when building can occur leads to higher risk of failure.
- Likelihood of Success is:

$$\prod_{q=1}^{Q} \delta^T = \delta^{QT}$$

Developer's Proposal Decision

- Project of size Q with profits $\pi(Q)$ has expected return:

$$\beta^T \delta^{QT} \pi(Q)$$

- Risk of failure adjusts discount factor from:

$$\beta \to \beta\delta^Q$$

- The first order condition with respect to Q is:

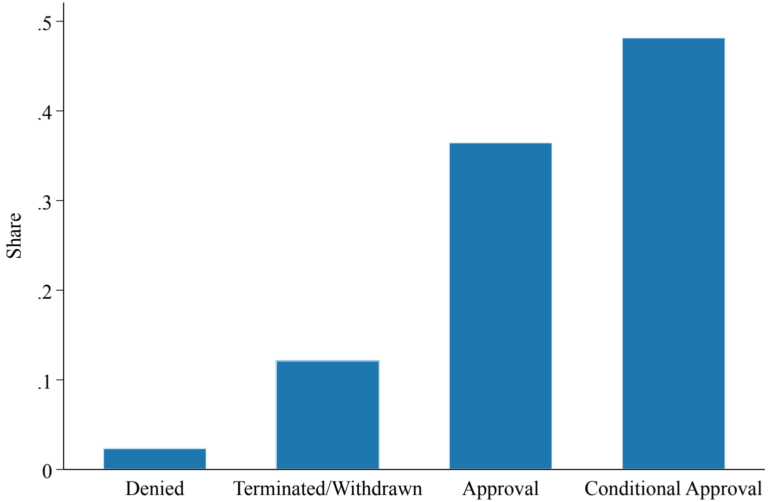
$$\beta^T \delta^{QT} T \ln(\delta)\pi(Q) + \beta^T \delta^{QT} \pi'(Q) = 0$$

$$T \ln(\delta)\pi(Q) + \pi'(Q) = 0$$

- Long approval timeline shifts projects off ideal FOC: $\pi'(Q) = 0$

Entitlement Pipeline: Facts

Entitlements Rarely Outright Denied



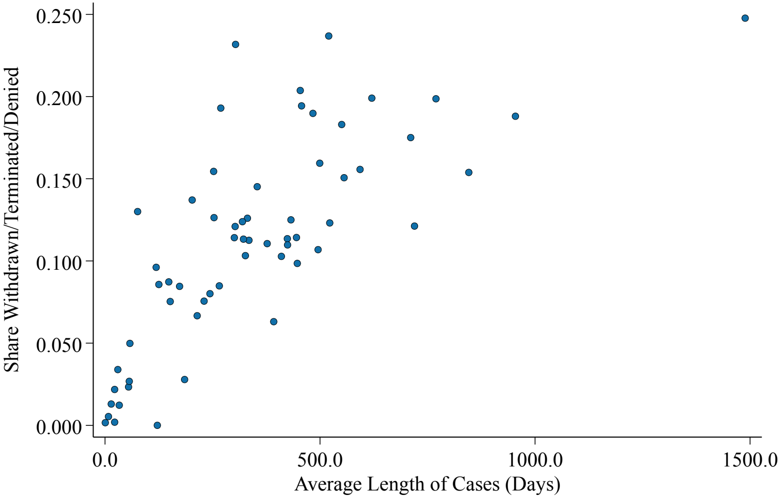
Not All Entitlements Equal: Ten Most Common

Suffix	N Cases	Avg. Length (days)	Appeal Rate	Frac. w/ Conditions	Frac. Withdrawn/Denied
CUB	3,598	235	0.046	0.623	0.076
SPP	2,930	358	0.077	0.789	0.114
WTF	2,498	1	0.000	0.000	0.002
TOC	2,397	252	0.053	0.284	0.126
CWC	1,334	21	0.000	0.001	0.020
HCA	1,284	424	0.080	0.562	0.114
VHCA	1,185	119	0.029	0.088	0.096
CDP	1,115	421	0.087	0.674	0.105
CU	1,030	434	0.136	0.620	0.134
AHRF	1,027	125	0.000	0.000	0.086

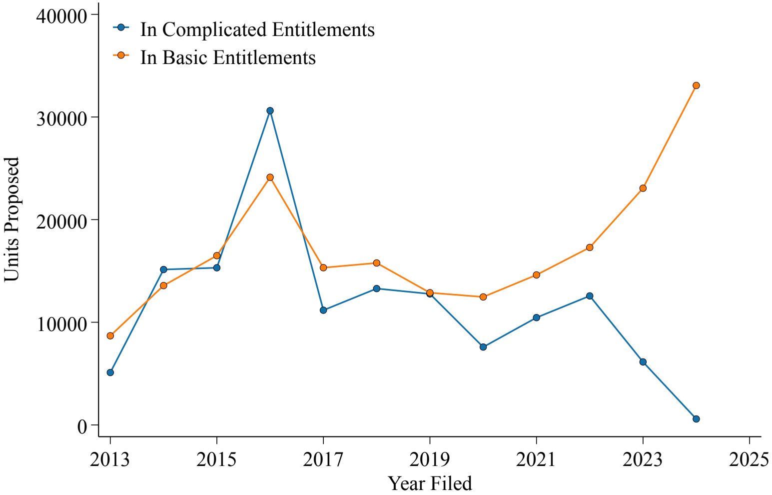
Not All Entitlements Equal: Ten Longest

Suffix	N Cases	Avg. Length (days)	Appeal Rate	Frac. w/ Conditions	Frac. Withdrawn/Denied
HD	158	930	0.297	0.468	0.165
VZC	143	868	0.259	0.510	0.161
MCUP	182	800	0.236	0.566	0.126
GPA	165	650	0.236	0.467	0.158
ZAD	767	617	0.111	0.585	0.202
ZC	308	548	0.104	0.331	0.133
SPR	909	546	0.250	0.649	0.136
ZAA	884	489	0.085	0.602	0.181
PMLA	659	463	0.067	0.524	0.115
ZV	972	458	0.098	0.608	0.189

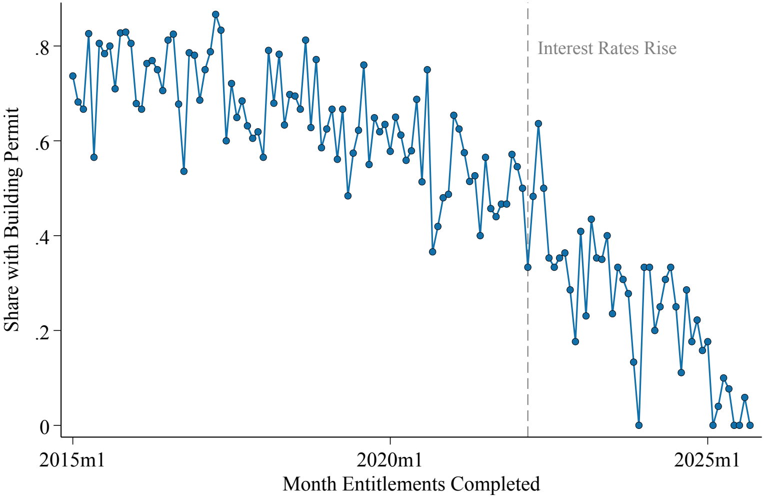
Longer Cases Have More Withdrawals



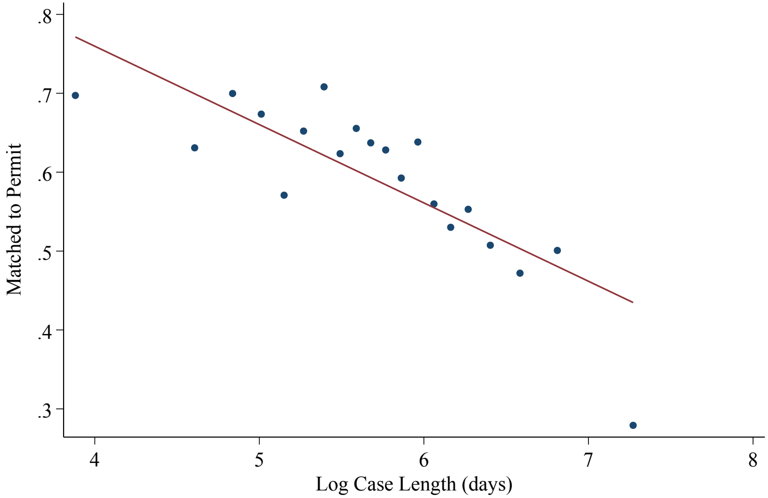
Share of Units in “Complicated” Entitlements Falling



Share of Entitlements Built Over Time



Longer Approved Projects More Likely to Be Unbuilt

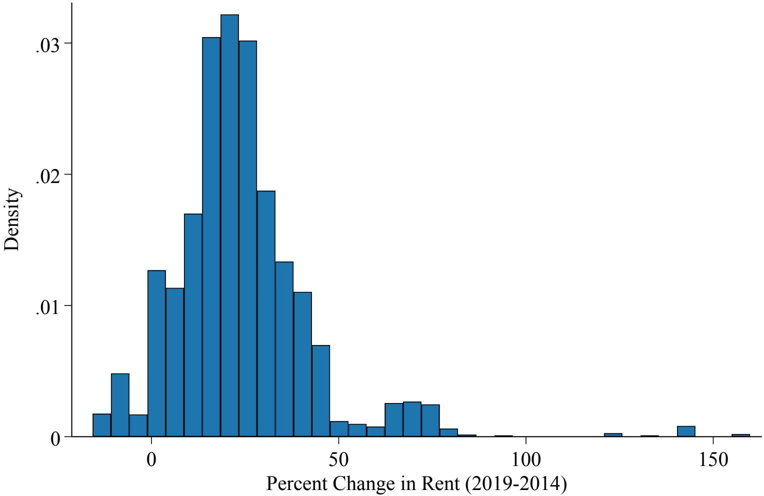


Regression Estimates

	Dep. Var: Matched to Permit			
	(1)	(2)	(3)	(4)
Log Units Proposed	-0.0208***		-0.0229***	-0.0273***
	(0.00366)		(0.00362)	(0.00554)
Time Trend	-0.00751***	-0.00676***	-0.00621***	-0.00646***
	(0.00208)	(0.00207)	(0.00207)	(0.00238)
Log Case Length (days)		-0.0912***	-0.0918***	-0.0906***
		(0.00854)	(0.00851)	(0.0125)
Log Total Hold Days				-0.0176**
				(0.00753)
No Hold Days				-0.271**
				(0.109)
Imp. Value Ratio				-0.0404
				(0.0369)
Assessed Land Value				-4.31e-09
				(4.75e-09)
Log Median Rent in Tract (2014)				0.0000567
				(0.0578)
Log Renter Share in Tract (2014)				-0.00522
				(0.0201)
Log Median HH Income in Tract				0.0157
				(0.0373)
Rent Growth in Tract (2014-2019)				-0.0522
				(0.0439)
Observations	4289	4226	4210	3105
R^2	0.178	0.198	0.207	0.241

Standard errors in parentheses

Nominal Rent Growth Positive



Lessons

- Large and long approval projects less likely to be completed: hard to distinguish whether it's something about the approval process vs something inherent to the project
  - Ideal: instrument for time in approval process using congestion—likely to be underpowered
- Rent growth a negative predictor of entitlement completion. Two stories
  - Reverse causality: completed entitlements are a shift in supply and lower rents
  - High upfront costs: little speculative behavior $\implies$ potentially missing projects
- While rents are weak predictors of building, time series suggests cost increases are very important
- Next Steps: hazard model, continue work on predicting profits at proposal, use that to see if observably speculative behavior (projects unlikely to pencil today but would in the future), write.

Mechanism: Political Risk

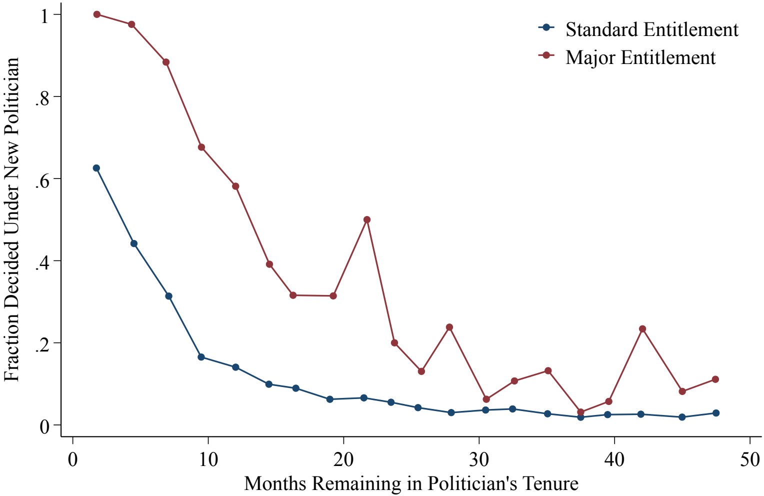
Lessons

- Story: many stakeholders and long timelines create risk of one stakeholder failing
- Observable mechanism: council member or mayor championing the project may leave office
- LA has member deference like in Chicago, NYC but only 15 Council Districts
- Some suggestive evidence that developers propose complicated projects with confidence over decisionmaker

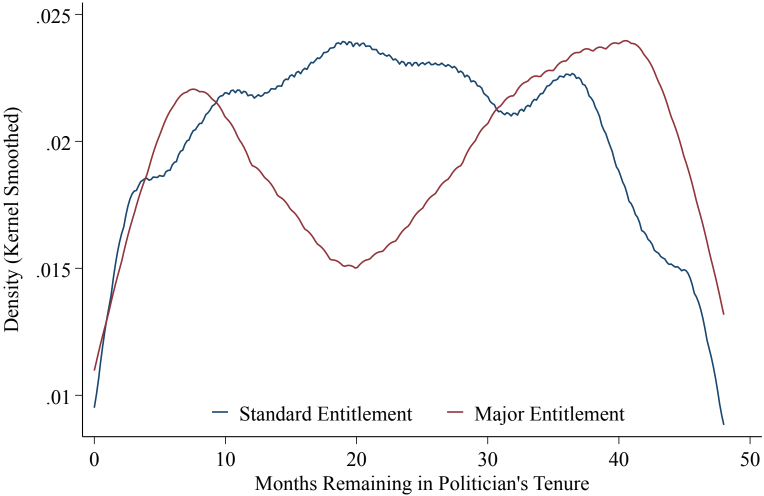
Council Members in District 4

Vacant		April 17, 2001 – November 1, 2001	
 **Tom LaBonge** (Silver Lake)	Democratic	November 1, 2001 – June 30, 2015	Elected to finish Ferraro's term. Re-elected in 2003. Re-elected in 2007. Re-elected in 2011. Retired.
 **David Ryu** (East Hollywood)	Democratic	July 1, 2015 – December 14, 2020	Elected in 2015. Lost re-election.
 **Nithya Raman** (Silver Lake)	Democratic	December 14, 2020 – present	Elected in 2020. Re-elected in 2024.

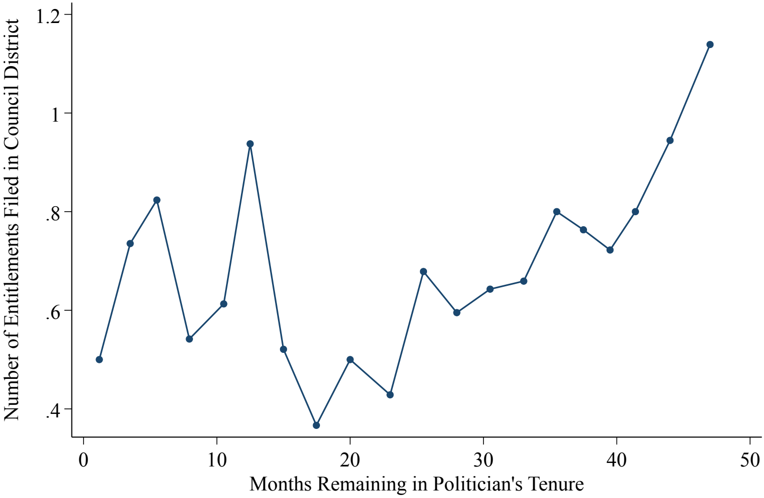
Rising Chance of New Council Member Near End of Tenure



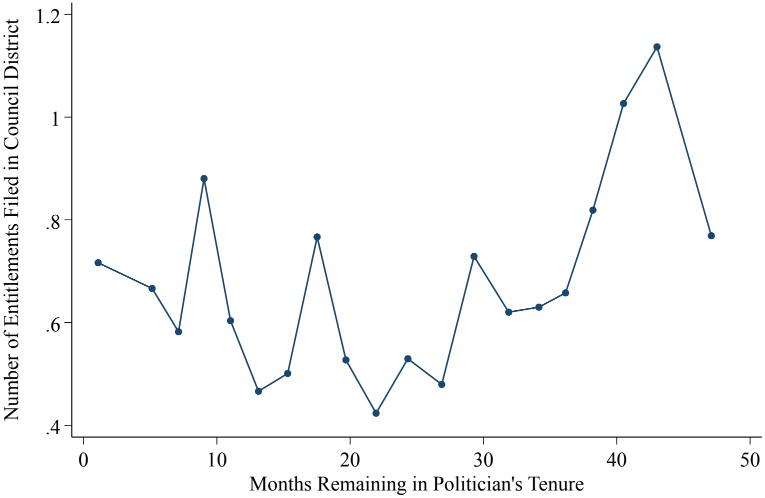
Major Entitlements Proposed Bimodally



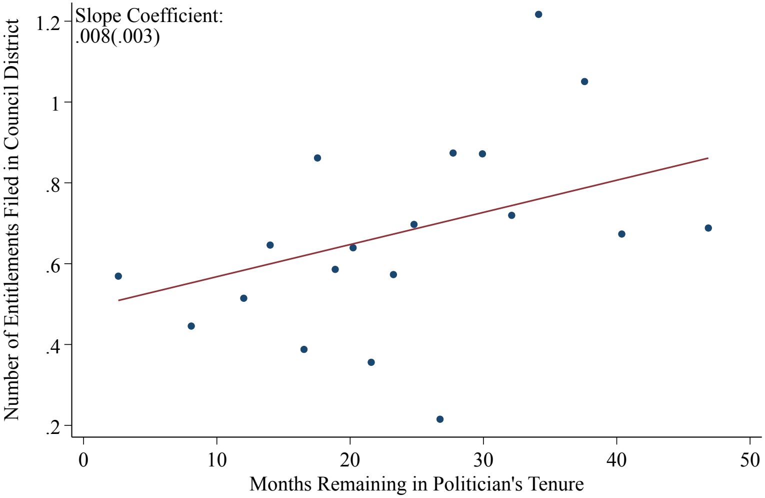
Major Entitlements At District Level



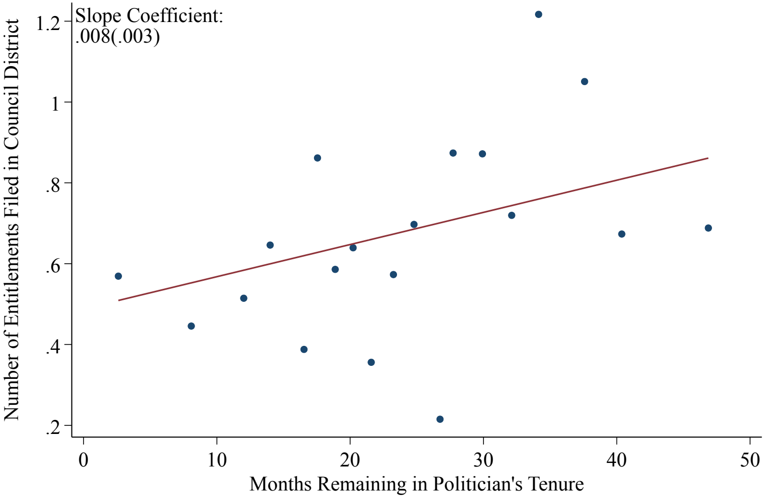
Major Entitlements At District Level: CD FEs



Major Entitlements At District Level: CD+Monthly FEs



Half as Many Major Projects at End of Tenure



Sample Issues

- Analysis in LA limited by 15 districts and four year terms
- Parcel analysis could be more precise
- Could supplement with additional data:
    1. Land Use Changes in NYC: have data in hand
    2. Chicago Planned-Developments/Zoning Map Changes: connection at planning department but not sure if worth reaching out.

NYC Learning by Doing

Sample Issues

- Analysis in LA limited by 15 districts and four year terms
- Parcel analysis could be more precise: complementary to above entitlements analysis
- Could supplement with additional data:
  1. Land Use Changes in NYC: have data in hand
  2. Chicago Planned-Developments/Zoning Map Changes: connection at planning department but not sure if worth reaching out.